

CHONGJIN, Korea, Democratic People's Republic of

WMO: 470080

Lat: 41.783N

Lon: 129.817E

Elev: 43

StdP: 100.81

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -12.3	(c) -10.8	(d) -26.2	(e) 0.3	(f) -8.5	(g) -23.4	(h) 0.5	(i) -8.0	(j) 9.8	(k) -7.1	(l) 8.6	(m) -8.0	(n) 2.9	(o) 320	(p) 0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.0	(c) 27.7	(d) 22.9	(e) 26.4	(f) 22.3	(g) 25.3	(h) 21.9	(i) 24.3	(j) 26.4	(k) 23.5	(l) 25.5	(m) 22.7	(n) 24.6	(o) 1.7	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	18.6	25.8	22.9	17.7	24.9	22.1	16.8	24.2	74.0	26.7	70.8	25.4	67.7	24.5	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.5	6.1	4.9	-15.2	30.9	2.2	1.4	-16.8	31.9	-18.1	32.7	-19.4	33.5	-21.0	34.5
			-16.4	25.6	1.9	1.4	-17.7	26.7	-18.8	27.5	-19.9	28.3	-21.2	29.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	-4.9	-3.1	1.8	7.9	12.3	16.6	20.8	22.3	18.0	11.8	3.8	-2.3
	DBStd	9.75	3.09	3.24	3.39	3.29	3.22	2.91	2.82	2.14	2.41	3.11	4.06	3.46
	HDD10.0	1768	463	367	255	81	13	0	0	0	0	18	189	383
	HDD18.3	3727	721	601	513	314	188	67	9	1	34	202	437	641
	CDD10.0	1334	0	0	1	16	86	199	336	381	239	74	2	0
	CDD18.3	251	0	0	0	0	2	16	87	123	23	0	0	0
	CDH23.3	857	0	0	0	0	8	53	296	454	46	0	0	0
	CDH26.7	95	0	0	0	0	1	6	36	49	3	0	0	0
Wind	WSAvg	1.4	1.6	1.6	1.6	1.4	1.3	1.1	1.0	1.2	1.3	1.6	1.7	1.7
Precipitation	PrecAvg	643	10	9	14	44	68	87	141	127	88	35	26	16
	PrecMax	1028	39	44	48	222	236	218	286	458	239	92	92	70
	PrecMin	457	1	0	0	0	0	8	24	9	9	3	0	1
	PrecStd	131	10	11	13	45	52	50	75	86	65	29	25	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	7.1	14.4	20.8	24.3	27.4	29.6	29.8	26.8	20.9	15.2	7.8
		MCWB	1.0	3.0	7.9	11.7	15.2	19.7	24.0	24.1	20.0	15.2	10.5	4.9
	2%	DB	2.2	4.8	11.0	17.4	21.4	24.7	27.5	28.0	24.6	19.5	13.3	5.7
		MCWB	-0.6	1.6	5.7	10.4	14.2	18.4	23.0	23.6	19.6	15.1	9.5	3.0
	5%	DB	0.9	3.2	8.8	15.0	19.4	22.8	26.2	26.8	23.3	18.3	11.6	4.2
		MCWB	-1.4	0.6	4.4	9.1	13.3	17.8	22.5	23.0	19.0	14.2	8.1	1.5
	10%	DB	-0.3	1.9	7.0	13.1	17.5	21.0	25.0	25.7	22.2	17.1	10.0	2.7
		MCWB	-2.3	-0.4	3.4	8.1	12.4	17.2	22.3	22.5	18.4	13.4	6.9	0.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	3.6	8.4	12.9	16.7	21.2	25.1	25.7	22.6	17.9	11.6	5.4
		MCDB	3.5	6.0	13.9	18.4	21.7	25.3	27.6	27.8	24.8	19.7	14.1	7.1
	2%	WB	-0.1	2.1	6.2	11.1	15.1	19.8	24.1	24.7	21.0	16.2	10.1	3.5
		MCDB	1.4	4.2	10.1	16.2	20.0	22.8	26.3	26.8	23.4	18.3	12.7	5.4
	5%	WB	-1.1	0.9	4.9	9.8	14.1	18.7	23.4	24.0	20.1	15.2	8.7	1.9
		MCDB	0.5	2.9	8.2	14.2	18.0	21.4	25.4	26.0	22.3	17.5	11.1	3.8
	10%	WB	-2.1	-0.2	3.7	8.6	13.1	17.8	22.6	23.2	19.2	14.0	7.3	0.6
		MCDB	-0.5	1.7	6.6	12.4	16.5	20.3	24.5	25.0	21.3	16.5	9.6	2.4
Mean Daily Temperature Range	5% DB	MDBR	6.5	6.6	6.6	7.1	6.5	5.0	4.3	5.0	7.1	7.9	7.0	6.5
		MCDBR	7.5	7.8	9.1	10.1	10.5	8.3	6.7	6.4	7.9	8.7	9.1	8.0
		MCWBR	6.4	6.4	6.0	5.7	5.4	4.3	3.7	3.8	4.9	6.4	7.3	6.7
	5% WB	MCDBR	7.2	7.5	8.5	9.4	8.9	6.5	5.6	5.3	6.5	7.9	8.7	7.7
		MCWBR	6.4	6.4	6.3	5.9	5.2	3.9	3.5	3.6	4.5	6.3	7.3	6.9
Clear-Sky Solar Irradiance	taub	0.337	0.390	0.460	0.540	0.578	0.583	0.540	0.493	0.438	0.434	0.390	0.347	
	taud	2.176	2.048	1.909	1.780	1.777	1.851	2.038	2.168	2.234	2.101	2.122	2.155	
	Ebn at Noon	780	787	772	740	724	722	750	775	792	732	705	726	
	Edh at Noon	100	133	173	211	217	202	166	142	124	125	106	94	
All-Sky Solar Radiation	RadAvg	2.24	3.20	4.30	4.88	5.02	4.81	4.40	4.46	4.21	3.35	2.23	1.90	
	RadStd	0.14	0.21	0.28	0.34	0.51	0.63	0.55	0.34	0.38	0.24	0.28	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	-2.30	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	-1.08	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page